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Applied Data Mining

Assignment 3

Lines 19-66

The screenshot displays the Posit Cloud interface for a workspace named 'mod5'. The main editor shows R code for text mining, including loading the SnowballC library, removing stopwords, and calculating term frequency-inverse document frequency (tf-idf) weights. The console output shows the results of the `inspect(tfidf)` function, including the number of terms and documents, sparsity, and a matrix of term frequencies.

```
## R Code Snippet
stopwords("english")
### Table 20.5
# stopwords
library(SnowballC)
corp <- tm_map(corp, removeWords, stopwords("english"))
# stemming
corp <- tm_map(corp, stemDocument)
tdm <- TermDocumentMatrix(corp)
inspect(tdm)
### Table 20.6
tfidf <- weightTfidf(tdm)
inspect(tfidf)
```

Console Output:

```
<<TermDocumentMatrix (terms: 5, documents: 4)>>
Non-/sparse entries: 4/16
Sparsity           : 80%
Maximal term length: 7
Weighting          : term frequency - inverse document frequency (normalized) (tf-idf)
Sample            :
  Terms Docs
first  1 0 0 0
forth  0 0 0 1
second 0 1 0 0
sentenc 0 0 0 0
third  0 0 1 0
```

Environment Panel:

Package	Description	Source	Version
base	The R Base Package	Base	4.5.2
base	The R Base Package	Base	4.5.3
BH	Boost C++ Header Files	PPM [default/latest]	1.90.0-1
boot	Bootstrap Functions	CRAN	1.3-32
boot	Bootstrap Functions	CRAN	1.3-32
caret	Classification and Regression Training	PPM [default/latest]	7.0-1
class	Functions for Classification	CRAN	7.3-23
class	Functions for Classification	CRAN	7.3-23
cli	Helpers for Developing Command Line I	PPM [default/latest]	3.6.5
clock	Date-Time Types and Tools	PPM [default/latest]	0.7.4
cluster	"Finding Groups in Data": Cluster Analys	CRAN	2.1.8.1
cluster	"Finding Groups in Data": Cluster Analys	CRAN	2.1.8.2
codetools	Code Analysis Tools for R	CRAN	0.2-20
codetools	Code Analysis Tools for R	CRAN	0.2-20
compiler	The R Compiler Package	Base	4.5.2
compiler	The R Compiler Package	Base	4.5.3
cpp11	A C++11 Interface for R's C Interface	PPM [default/latest]	0.5.3
data.table	Extension of "data.frame"	PPM [default/latest]	1.18.2.1
datasets	The R Datasets Package	Base	4.5.2
datasets	The R Datasets Package	Base	4.5.3
diagram	Functions for Visualising Simple Graphs	PPM [default/latest]	1.6.5
digest	Create Compact Hash Digests of R Objects	PPM [default/latest]	0.6.39
dplyr	A Grammar of Data Manipulation	PPM [default/latest]	1.2.0

```

> ##### Table 20.2
>
> library(tm)
>
> text <- c("this is the first sentence!!",
+          "this is a second Sentence :)",
+          "the third sentence, is here",
+          "forth of all sentences")
> corp <- Corpus(VectorSource(text))
> tdm <- TermDocumentMatrix(corp)
> inspect(tdm)
<<TermDocumentMatrix (terms: 12, documents: 4)>>
Non-/sparse entries: 14/34
Sparsity : 71%
Maximal term length: 10
Weighting : term frequency (tf)
Sample :
      Docs
Terms 1 2 3 4
all    0 0 0 1
first  1 0 0 0
here   0 0 1 0
second 0 1 0 0
sentence 0 1 0 0
sentence, 0 0 1 0
sentence!! 1 0 0 0
the    1 0 1 0
third  0 0 1 0
this   1 1 0 0

```

```

> ##### Table 20.3

```

```

> # tokenization
> corp <- tm_map(corp, stripWhitespace)

```

Warning message:

```

In tm_map.SimpleCorpus(corp, stripWhitespace) :
  transformation drops documents

```

```

> corp <- tm_map(corp, removePunctuation)

```

Warning message:

```

In tm_map.SimpleCorpus(corp, removePunctuation) :
  transformation drops documents

```

```

> tdm <- TermDocumentMatrix(corp)
> inspect(tdm)
<<TermDocumentMatrix (terms: 10, documents: 4)>>
Non-/sparse entries: 14/26
Sparsity : 65%
Maximal term length: 9
Weighting : term frequency (tf)
Sample :
      Docs
Terms 1 2 3 4
all    0 0 0 1
first  1 0 0 0
forth   0 0 0 1
here   0 0 1 0
second 0 1 0 0
sentence 1 1 1 0
sentences 0 0 0 1
the    1 0 1 0
third  0 0 1 0
this   1 1 0 0

```

```
> ##### Table 20.4
>
> stopwords("english")
[1] "i" "me" "my" "myself" "we" "our" "ours" "ourselves"
[9] "you" "your" "yours" "yourself" "yourselves" "he" "him" "his"
[17] "himself" "she" "her" "hers" "herself" "it" "its" "itself"
[25] "they" "them" "their" "theirs" "themselves" "what" "which" "who"
[33] "whom" "this" "that" "these" "those" "am" "is" "are"
[41] "was" "were" "be" "been" "being" "have" "has" "had"
[49] "having" "do" "does" "did" "doing" "would" "should" "could"
[57] "ought" "i'm" "you're" "he's" "she's" "it's" "we're" "they're"
[65] "i've" "you've" "we've" "they've" "i'd" "you'd" "he'd" "she'd"
[73] "we'd" "they'd" "i'll" "you'll" "he'll" "she'll" "we'll" "they'll"
[81] "isn't" "aren't" "wasn't" "weren't" "hasn't" "haven't" "hadn't" "doesn't"
[89] "don't" "didn't" "won't" "wouldn't" "shan't" "shouldn't" "can't" "cannot"
[97] "couldn't" "mustn't" "let's" "that's" "who's" "what's" "here's" "there's"
[105] "when's" "where's" "why's" "how's" "a" "an" "the" "and"
[113] "but" "if" "or" "because" "as" "until" "while" "of"
[121] "at" "by" "for" "with" "about" "against" "between" "into"
[129] "through" "during" "before" "after" "above" "below" "to" "from"
[137] "up" "down" "in" "out" "on" "off" "over" "under"
[145] "again" "further" "then" "once" "here" "there" "when" "where"
[153] "why" "how" "all" "any" "both" "each" "few" "more"
[161] "most" "other" "some" "such" "no" "nor" "not" "only"
[169] "own" "same" "so" "than" "too" "very"
```

```
> ##### Table 20.5
>
> # stopwords
> library(SnowballC)
> corp <- tm_map(corp, removeWords, stopwords("english"))
```

```
Warning message:
In tm_map.SimpleCorpus(corp, removeWords, stopwords("english")) :
  transformation drops documents
```

```
>
> # stemming
> corp <- tm_map(corp, stemDocument)
```

```
Warning message:
In tm_map.SimpleCorpus(corp, stemDocument) : transformation drops documents
```

```
>
> tdm <- TermDocumentMatrix(corp)
> inspect(tdm)
<<TermDocumentMatrix (terms: 5, documents: 4)>>
Non-/sparse entries: 8/12
Sparsity : 60%
Maximal term length: 7
Weighting : term frequency (tf)
Sample :
      Docs
Terms 1 2 3 4
first  1 0 0 0
forth  0 0 0 1
second 0 1 0 0
sentenc 1 1 1 1
third  0 0 1 0
```

```

> #### Table 20.6
>
> tfidf <- weightTfIdf(tdm)
> inspect(tfidf)
<<TermDocumentMatrix (terms: 5, documents: 4)>>
Non-/sparse entries: 4/16
Sparsity          : 80%
Maximal term length: 7
Weighting         : term frequency - inverse document frequency (normalized) (tf-idf)
Sample           :
      Docs
Terms  1 2 3 4
first  1 0 0 0
forth  0 0 0 1
second 0 1 0 0
sentenc 0 0 0 0
third  0 0 1 0

```

Explanation:

The data was first converted into a corpus and represented as a Term-Document Matrix (TDM), where rows correspond to terms and columns correspond to documents. The initial matrix included all words which appeared in the text but created multiple instances of the same word because of different punctuation and formatting. The process began with preprocessing steps which involved the elimination of all unnecessary spaces and punctuation marks.

The process standardized the text while reducing redundant elements which resulted in a decrease of unique terms and better document consistency. The researchers created a list of common English stopwords which included words like "the" and "is" and "and" because these words lack important meaning. The process of stemming resulted in the transformation of words to their fundamental roots which included the conversion of "sentence" and "sentences" to "sentenc" which made the dataset less complex.

The matrix received a TF-IDF (Term Frequency–Inverse Document Frequency) weighting system which was implemented as the final step. The method establishes greater value for words which are present in only one document while it grants lesser value to words which are found throughout all documents. Common terms such as "sentenc" received minimal or nonexistent weight while uncommon terms maintained greater value.

